Graphic Era HILL UNIVERSITY

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University under section 2(f) of UGC Act, 1956

End Term (Odd) Semester Examination December 2025

Roll no.....

Name of the Course: B. Tech. CSE
Semester: IIIrd
Name of the Paper: Introduction to Cryptography
Paper Code: TCS-392

Time: 3 hours **Maximum Marks: 100**

Note:

(i) All questions are compulsory

(ii) Answer any two sub questions from a, b and c in each main question.

(iii) Total marks for each question is 20 (Twenty).

(iv) Each sub-question carries 10 marks.

Q1. (2×10 = 20Marks)

a. With the help of a neat and clearly labelled diagram, illustrate the OSI Security Architecture and explain the role of each component in detail. [CO1]

b. Critically examine the strengths and weaknesses of Classical Ciphers against modern cryptanalysis techniques . [CO1]

c. Explain the principles of symmetric encryption with suitable block diagram. [CO1]

Q2. (2×10 = 20Marks)

a. Perform frequency analysis on the ciphertext "ZOLSSP" and attempt its decryption using a monoalphabetic substitution technique. Show all intermediate steps.[CO2]

b. Encrypt the plaintext "1101 0110" using a single round of simplified DES (S-DES). Show all steps. [CO2]

c. Discuss the necessity of Random Numbers and Pseudorandom Numbers in cryptographic systems. Provide suitable real-world applications for each. [CO2]

Q3. (2×10 = 20Marks)

a. Explain the importance of mathematical background in cryptography. Discuss with reference to prime numbers and modular arithmetic. [CO3]

b. A company aims to reduce the frequency of rekeying operations for secure communication channels. Recommend an appropriate key distribution mechanism and justify your selection [CO3]

c. Critically examine the role of message authentication codes (MACs) in ensuring integrity. [CO3]

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Q4. (2×10 = 20Marks)

a. Given p=11, Choose a public exponent e=7.

(a) Compute the public key (n,e)and the private exponent d.

(b) Encrypt the message M=8.

(c) Decrypt the resulting ciphertext to recover M. [CO4]

b. Compare Symmetric Encryption, Asymmetric Encryption, and Hybrid Encryption. Recommend an optimal strategy for secure mobile banking communication and justify your choice. [CO4]

c. Discuss the challenges associated with Key Distribution in large-scale networks and suggest practical solutions to overcome these issues. [CO4]

Q5. (2×10 = 20Marks)

a. Discuss in detail the ethical and legal aspects of cybercrime, focusing on Intellectual Property Rights and Privacy. [CO5]

b. A company's network is under frequent intrusion attempts. Design a basic intrusion detection strategy for them. [CO5]

c. Propose a complete Network Plan for an organization facing frequent malware infections. Discuss measures at OS level, network level, application level, and user-training level. [CO5]

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